

Explosive Shoulder Strength: A Dominant Predictor of Women's Volleyball Playing Ability

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ABSTRACT

Considering the correlation between strength and women's volleyball playing ability, this paper intends to inquire the possibilities to find out best test for explosive shoulder strength. The purposive sampling technique was used for sixty volleyball players from northern region of India. Data was collected during pre-national coaching camps and during volleyball school national championship. The measurement tools for shoulder strength were namely overhead medicine ball throw, basketball throw and softball throw. The volleyball playing ability was calculated by self made five point rating scale. This scale was implemented by a panel of three qualified judges, having adequate understanding of volleyball game. The results of experiment revealed that volleyball playing ability of women and overhead medicine ball throw; basketball throw and softball throw have strong, stronger and strongest correlation respectively. The level of significance for experiment was 0.01. The sports application of present research is for all the stakeholders, those who are willing to achieve the spectacular performance.

Keywords:

explosive strength, predictor, muscular effort, volleyball playing ability.

INTRODUCTION

Modern era of volleyball is full of competition and there is always a struggle for achieving supremacy and excellence. Today, the country's overall superiority at the international fore is reflected through sports performance and interlinked with national pride and culture. There is a need of consistent efforts on the part of researchers to find out best volleyball talent and performance predictors. Exploring the requirements of volleyball Chander (2010) argued that it is a sport that not

only requires a number of individual skills and technical abilities in each player, but also requires that the players have a high degree of physical and physiological considerations in order to be successful. Further, Chander and Mukherjee (2012) emphasized that for exceptional volleyball player sustaining level of strength and power is required.

Bompa and Carrera (2005) highlighted the requirements of volleyball players and listed qualities like quick reaction, explosively take off for attacking, blocking and diving actions.

Further, they added the qualities of maximum strength, power and special endurance required during prolonged competitive period. The competitive volleyball involves strength along with other fitness components for maintenance of mental and physical stress throughout the game (Rao, 2003). Morrow et al. (1979) concluded that the fundamental attributes of speed and strength are prerequisites of successful performance in volleyball. Lehnert, Lamrová and Elfmark (2009) illustrated a need of explosive power and indicated that it is most important part of skillful playing in volleyball. The performance in majority of sports needs speed and power (Faigenbaum et al. 2009). Keeping in view the demands of explosive strength/ power, the question now comes is how to improve the strength? Research in this regard says, that resistance training in teenagers leads to improvement in muscular strength along with power and thereby improves sports performance (Annesi, Westcott, Faigenbaum and Unruh, 2005; American Academy of Pediatrics, et al. 2008; Faigenbaum et al. 2009).

Most of sports movements fundamentally require muscle power and force generation (Thibaudeau, 2006; Kraemer and Newton, 1994; Harries, Lubans and Callister, 2012). Zatsiorsky and Kraemer (2006) explained strength as an ability to resist the external force with musclecontraction. Further Kurt et al. (1997) argued that strength as ability

of a muscle to exert force; the maximum amount of effort that muscle can apply in a contraction. Looking in to the scope of present paper, muscle strength may be described as muscles contraction used for starting or stopping of a movement. Another form of strength that is required for playing volleyball is explosive strength or power. It is defined as the ability to apply the maximal force in lesser time. (Zatsiorsky and Kraemer 2006). Kurt et al (1997) defined power as, the rate at which work is performed (work/time), a combination of strength and velocity. Concluding the concept of explosive power, it is an ability to perform a maximum effort in minimum time.

METHODOLOGY

SAMPLING

The participants who acted as sample for this experiment were sixty women volleyball players of under-nineteen age group from northern India. They were sports person, representing six participating units of northern Indian states. Purposive sampling technique was used for collection of data. Sampling procedure for shoulder strength tests was completed during pre-national coaching camps. The volleyball playing ability was judged during volleyball championship, celebrated under the aegis of "School Games Federation of India" held at Dabra (Madhya Pradesh), India. It was observed on a five point rating scale by the panel of three experienced and

qualified judges having sufficient experience in the field of volleyball.

SELECTION OF TOOL

The tools for power test of shoulder strength were selected on the advice of coaches' and experienced physical education teachers. Researcher's experience, related literature review, feasibility of the tests, equipments accessibility and acceptability of the subjects were also taken care. The aim of testing explosive shoulder strength was to discover best

test/ tool for identifying the predictor for players' volleyball playing ability. Taking into consideration all the above factors, overhead medicine ball throw, basketball throw and softball throw were finalized for testing explosive shoulder strength. The criterion-measure chosen for this study was overall volleyball playing ability.

DATA RELIABILITY

The reliability of tools was established with the help of subjects' reliability.

Table (a), Reliability coefficient of test-retest score of explosive shoulder strength test

Sr. No.	Test Variables	Reliability
i.	Two Hand Overhead Medicine Ball Throw	.85
ii.	Two Hand Basketball Throw	.82
iii.	One Hand Softball Throw	.89

A high correlation of reliability in 'test and the retest' scores of explosive shoulder strength was found.

VOLLEYBALL PLAYING ABILITY

The volleyball playing ability was observed/ evaluated during national championship. This ability was observed

on five point rating scale by a panel of three experienced and qualified experts having sufficient experience in the game. The following guidelines were provided to the judges for rating the volleyball playing ability:

Table (b) Volleyball Playing Ability Scale

Serial Number	Guiding Instructions	Score Points
i.	Excellent in all aspect of the game	5
ii.	Very good in all aspect of the game	4
iii.	Good in all aspect of the game	3
iv.	Above average in all aspect of the game	2
v.	Average in all aspect of the game	1

The rating scale, used for observation of playing ability is given in table (b). The ratings were applied keeping in view of skill proficiency; anticipation and quick reaction; tactics and strategies interactions; knowledge of rules and games; ability to change tactics when new situation demands; ability to analyze opponent's movement and respond accordingly and to consider the overall rational of the game.

ADMINISTRATION AND DATA COLLECTION

All the volleyball players, who acted as sample, were tested during pre-national volleyball coaching camps and during national championship. The data was collected by administering selected explosive strength tests, namely two hand overhead medicine ball throw, two hand basketball throw and one hand softball throw. The administration of overhead medicine ball throw was completed by making players stand behind the throwing line and keeping the medicine ball behind the shoulder and then throw it without taking step on or across the throwing line. Three consecutive throws were permitted, and the best throw was recorded the nearest centimeter as score. Second, the basketball throw, was included for the measurement of shoulder girdle strength and coordination. The subject stood behind the throwing line and used technique of throwing as per her wishes. She had to throw down the basketball to a distance. She must not step on or cross

the throwing line. Three consecutive throws were permitted, and the best throw was recorded to the nearest feet. Third, the softball throw, in which the subject stood behind the throwing line and throws the softball as per her wishes. Three consecutive throws were permitted, and the best throw was recorded to nearest feet. The time of all testing scheduled at the point when impact of their training and conditioning had reached the top performance. The chance of practice before the administration of test was given to players, so that they get acquainted with required testing. Further, the researcher had ensured uniform testing conditions by conducting tests during morning and evening training sessions.

RESULTS AND DISCUSSION

Data analysis is significant part for exploring data in consequential results. It organizes, interprets, structures and presents the data for meaningful context. The analysis of data was completed with use of Pearson's product moment coefficient of correlation and to see the connection among explosive shoulder strength and volleyball playing.

Descriptive Analysis of Shoulder Strength

The analysis of explosive shoulder strength among volleyball players containing mean, standard deviation, standard error of mean and coefficient of variation percentage has been illustrated in table (c):

Table (e) highlights the inter-relationship between overhead medicine ball throw, basketball throw and softball throw among north-Indian women volleyball players in terms of Pearson's product moment correlation coefficients. The outcomes revealed the positive relationship overhead medicine ball throw and basketball throw ($r =$

0.453) and it was found significant at 0.01 level of significance. The association between basketball throw and softball throw was ($r = 0.656$) and it is statistically significant at 0.01 level. The correlation among overhead medicine ball throw and softball throw ($r = 0.469$) was found statistically significant at the level of 0.01 level.

Table (f): Relationship between explosive shoulder strength and volleyball playing ability

S.N.	Explosive Shoulder Strength	Correlation coefficients	Level of significance
i.	Overhead Medicine Ball Throw	0.522	< .01
ii.	Basketball Throw	0.580	< .01
iii.	Softball Throw	0.690	< .01
Number of players=60, Degree of freedom = 58			

It is illustrated in table (f) that correlation of women volleyball playing ability with overhead medicine ball throw ($r = 0.522$), basketball throw ($r = 0.580$) and softball throw ($r = 0.690$) were found statistically significant at the level of 0.01, as calculated value is greater than the table value at 1% level of significance.

The correlation table (f) clearly indicates that all three explosive shoulder tests namely overhead medicine ball throw, basketball throw and softball throw have positive significant relationship with volleyball playing ability of women players. Further it also signify that softball throw have high correlation with playing ability in volleyball.

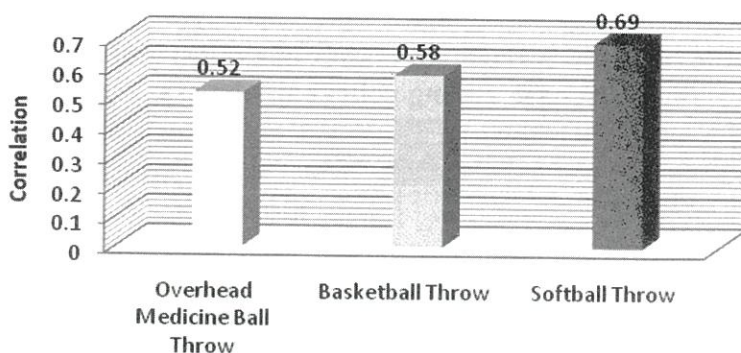


Figure (z): Chart showing the correlation between explosive shoulder strength and volleyball playing ability

Table (c): Descriptive analysis of explosive shoulder strength among women volleyball players

Sr. No	Name of Test	Unit	Minimum	Maximum	Mean	SD*	SEM*
i.	Overhead Medicine Ball Throw	—	411	870	630.14	98.43	11.91
ii.	Basketball Throw	Feet	33	57	43.38	5.98	.729
iii.	Softball Throw	Feet	65	133	99.27	14.84	1.87
N=60, * standard deviation, **standard error of mean							

It is evident from table (c) that overhead medicine ball throw had mean 630.14, 98.43 standard deviation, and 11.91 standard error of mean. Basketball throw had 43.38 mean, 5.98 standard deviation, and 0.729 standard error of mean. Softball throw had 99.27 mean, 14.84 standard deviation, and 1.87 standard error of mean.

Table (d): Descriptive statistics of volleyball playing ability among women players

Name of the Test	N	Range	Minimum	Maximum	Mean Static	Std. Error of Mean	Std. Deviation
Volleyball Playing Ability	60	4.00	1.00	5.00	3.763	.1354	1.0821

Table (d) indicates that volleyball playing ability among women volleyball players have 4.00 range, 3.763 static mean, .1354 standard deviation and 1.0821 standard error of mean.

Table (e): Inter-correlation matrix of explosive shoulder strength among volleyball players

Test	Overhead Medicine Ball Throw	Basketball Throw	Softball Throw
Overhead Medicine Ball Throw	1	0.453**	0.469**
Basketball Throw	0.453**	1	0.656**
Softball Throw	0.469**	0.656**	1
N=60, **Significant at the 0.01 level, Degree of freedom = 58			

Table (d) indicates that volleyball playing ability among women volleyball players have 4.00 range, 3.763 static mean, .1354 standard deviation and 1.0821 standard error of mean.

Figure (z) depicted fascinated information that although overhead medicine ball throw, basketball throw and softball throw through have positive relationship with volleyball playing ability, but in case of softball throw, it is very high. It can be concluded from this result that this test can be used for testing explosive shoulder strength among volleyball players for identification of talent. Further, it may be used a tool for predicting future volleyball performance. Chander (2018) also concluded that testing explosive shoulder through softball throw is also a predictor of vertical jumping ability.

Conclusion and Recommendations

This present paper has volleyball training and coaching application for all those stakeholders who are working for betterment of volleyball performance. Further, the inter-relationship between shoulder strength test and volleyball playing ability may help to develop better understanding of strength training

concept. After all careful deliberations, the researcher has following recommendation for volleyball coaches and trainers:

- i. Overhead medicine ball throw, basketball throw and softball throw have statistically significant correlation to volleyball playing ability.
- ii. Softball throw was found as best suitable for judging explosive shoulder strength for volleyball playing ability, subjected to availability of free throwing space. Especially, in hilly area or metro cities, where sufficient space is not available, one can use basketball throw or least predictive overhead medicine ball throw.
- iii. Keeping in view the nature of the volleyball game, it is recommended that while talent identification for future volleyball players, a player may be tested through softball throw. man Kinetics, p 28.

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